# Science

Nature Notebook
Nature Walks & Scouting
Astronomy Lessons
Astronomy Labs

Alveary

## Science: Astronomy

GRADE(S): 11-12

### About the Course

This course includes the following topic(s): Astronomy Lessons, Astronomy Labs, Nature Notebook: Grades 9-12, Nature Walks & Scouting: Grades 9-12

About Nature Notebook: Grades 9-12
Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

About Nature Walks & Scouting: Grades 9-12
Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

About Astronomy Lessons
An elective in Earth Science, Astronomy is a fascinating topic in which learners explore celestial objects, history, and philosophy. Teachers wishing for explicit instruction on topics related to world view should include the optional Crossroads text.

About Astronomy Labs
Labs/fieldwork are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

### Placement & Combining Tips

Nature Notebook: Grades 9-12
Learners may be combined and follow their own interests.

Nature Walks & Scouting: Grades 9-12
Learners may follow their own interests or follow the plan of their local scouting troop or natural history club.

Astronomy Lessons
Recommended for students in Grades 11-12, depending on math placement. Students should have completed Algebra 1 (Algebra 2 if taking Denison Success) and Geometry.

### Scheduling

| GRADE | SCHEDULE INFO. | BOOKS |
|---|---|---|
| 9-12 | Nature Notebook: Grades 9-12<br>1+ time/week 20 min+ | |
| 9-12 | Nature Walks & Scouting: Grades 9-12<br>1 time/week 30 min+ | |
| 11-12 | Astronomy Lessons<br>5 times/week 45 min | 100 Things to See in the Night Sky<br>The Last Stargazers<br>Astronomy Textbook<br>The Day Without Yesterday<br>The Crossroads of Science and Faith: Answers to Exercises<br>The Crossroads of Science and Faith: Astronomy Through a Christian Worldview |
| 11-12 | Astronomy Labs<br>1 time/week 60 min | |

Sample Weekly View

| Day 1 | Day 2 | Day 3 | Day 4 | Day 5 |
|---|---|---|---|---|
| **Science: Astronomy** | | | | |
| Astronomy Lessons<br>Nature Walks & Scouting: Grades 9-12 | Astronomy Lessons | Astronomy Lessons | Astronomy Lessons<br>Nature Notebook: Grades 9-12 | Astronomy Lessons<br>Astronomy Labs |

## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Astronomy Lessons
☐ Carve out time to continue or establish the regular habit of spending time in nature, including the use of a nature journal, as appropriate.

☐ Obtain materials from the supply lists.

☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.

☐ Bookmark or print Quick Links, as needed.

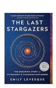
## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.
∞ View Book List Details

**Science: Astronomy**

### Astronomy Lessons

100 Things to See in the Night Sky



The Last Stargazers


Astronomy Textbook


The Day Without Yesterday


The Crossroads of Science and Faith: Answers to Exercises


The Crossroads of Science and Faith: Astronomy Through a Christian Worldview

## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.
∞ View Basic Supplies

(No Subject Supplies Assigned)

## Quick Links

Click THIS text or scan the QR code for links.

**Science: Astronomy**

### Astronomy Lessons

- SkyView Lite for iOS
- SkyView Lite for Android
- Stellarium
- Hubble Space Telescope
- Webb Space Telescope
- Earth Sky
- Astronomy Picture of the Day
- Veritas Forum
- Galaxy Zoo
- Globe at Night
- Exoplanet Watch
- Space Math
- Alveary Bookshelf
- Crossroads Answer Key Part 1
- Crossroads Answer Key Part 2

# Science: Astronomy

## How To Approach

### Introduce

- If starting a new book or a new topic in the book, then look at the title or a picture and take a moment to consider previous ideas and experiences.

- If continuing a previous reading, recap what was read previously. Often, the title of the book's section can help to draw out the main idea.

### Read

- Read or do, as instructed in the lessons, making note of or flagging unfamiliar terms, interesting ideas, important dates, inspiring quotes, etc.

- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

- As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.

- If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab/field work further supports major concepts from the text.

### Narrate

- Process the ideas of the lesson by retelling, defining a concept, explaining the links in a chain of thought, etc. Do this orally or silently to yourself.

- Use words, pictures, outlines, etc.

- If a particular idea cannot be narrated, then read or examine the text again.

### Discuss

- Consider with the teacher any thoughts, confusion, or concerns about the passage.

- If understanding is still uncertain, try rereading the passage or do some personal research on the topic.

### Connect

- Follow any extra links, examine any sidebars in the text, or pursue additional reading, depending on student interest.

# Science: Astronomy

## How To Approach Labs

### Introduce

• Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.

• Relevant concept(s) are introduced in the text.

• Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.

### Lab Procedure

• Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.

• Record all data and observations in the lab notebook.

### Analysis & Conclusions

• After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.

• If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

# Science: Astronomy

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## Term 1

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 1 ⬚ **45m Astronomy Lessons - Lesson 1**
*Identification*

▯ Materials: Astronomy Text, star finder

**→ INTRO**
This is your day each week to prepare for your field studies. You will begin with a short reading, but most of the lesson will be used for planning. To find a local club, explore your locale at the links below:
∞ Link: Night Sky Network
∞ Link: New Planetarium
∞ Link: Royal Astronomical Society

**→ READ, NARRATE, & DISCUSS**
Unit I Lesson 1 through the section "Shifting Skies" (to "in the sky.")

**→ PREP**
• Learn to orient yourself with the star finder.
• Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
• Note what time the sky becomes dark enough to see the stars.
• Check the weather and note which nights will have cloud cover so that you can plan a night each week to do some sky gazing.
• As you are able, seek out local observatories or clubs to increase your knowledge and experience locating objects in the sky.

WEEK 1 ⬚ **45m Astronomy Lessons - Lesson 2**
*How Astronomy Works, Highlights from History*

▯ Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit I Lsn 2. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 1 ⬚ **45m Astronomy Lessons - Lesson 3**
*Astronomy Literature*

▯ Materials: The Last Stargazers

**→ INTRO**
Two days each week will be spent reading important works of astronomical literature. The lesson plans give you an idea of how many pages to read each week to finish the book on schedule, but this is flexible. If a book takes longer than suggested, you might read outside the lesson time or even leave off without finishing it. If you read faster than suggested, use the extra time for other activities in the course.

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 1 ⬚ **45m Astronomy Lessons - Lesson 4**
*Astronomy Literature*

▯ Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

# Science: Astronomy

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## Term 1

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 1 ⬚ **45m Astronomy Lessons - Lesson 5**
*Astronomy Current Events and Research*

▯ Materials: computer, optional: Crossroads of Science and Faith text

**→ INTRO**
About half of these days will be spent reading current events; the other half will be spent participating in citizen science. Keep a log or journal entry of your findings and thoughts in your Astronomy Notebook.

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links).

WEEK 1 ⬚ **60m Astronomy Labs - Lesson 1**

▯ Materials: journal, star finder

**→ OBSERVE**
Simply notice on this first night of observing. What are your first impressions? Or are you able to relocate any objects previously found? Do you notice anything new based on your use of the star finder or Quick Links? Keep a record of what you observe.

WEEK 2 ⬚ **45m Astronomy Lessons - Lesson 6**
*Identification*

▯ Materials: 100 Things to See in the Night Sky

**→ READ, NARRATE, & DISCUSS**
How to Use This Book

**→ PREP**
- Learn to orient yourself by locating or reminding yourself where the four cardinal directions point in your place.
- Practice measuring objects at a distance using your hands.
- Review your viewing experience last week. Was it a good spot? Are there other possibilities that would be better? Can you do anything to improve your experience?
- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Check the weather and note which nights will have cloud cover so that you can plan a night each week to do some sky gazing.

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 2 ⬚ **45m Astronomy Lessons - Lesson 7**
⬚ *45m Astronomy - Lesson 7*

▯ Materials: Astronomy Text

How Astronomy Works, Gravity

**→ READ, NARRATE, & DISCUSS**

# Science: Astronomy

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## Term 1

Unit I Lsn 3. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

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WEEK 2

☐ **45m Astronomy Lessons - Lesson 8**

*Astronomy Literature*

Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**

Read about 19 pages per week (over 2 lesson days).

**→ WRITE**

Write a narration in your Astronomy Notebook.

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WEEK 2

☐ **45m Astronomy Lessons - Lesson 9**

*Astronomy Literature*

Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**

Read about 19 pages per week (over 2 lesson days).

**→ WRITE**

Write a narration in your Astronomy Notebook.

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WEEK 2

☐ **45m Astronomy Lessons - Lesson 10**

☐ *45m Astronomy - Lesson 10*

Materials: computer, optional: Crossroads of Science and Faith text

Astronomy Current Events and Research

**→ READ CURRENT EVENTS**

EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

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WEEK 2

☐ **60m Astronomy Labs - Lesson 2**

Materials: journal, star finder

**→ OBSERVE**

Notice any objects that are or are becoming familiar. In what directions do they currently lie? Using your hands, measure their locations in the sky. Do these locations change during the evening? Keep a record of what you observe.

★ STUDENT/TEACHER TIP
Note that next week's fieldwork should be done in one or more daytime sessions!

# Science: Astronomy

Click THIS text or scan the QR code for links.

## Term 1

WEEK 3 ⬚ **45m Astronomy Lessons - Lesson 11**
*Identification*

> ▯ Materials: 100 Things to See in the Night Sky, binoculars, paper, tripod/stand for binoculars

**→ READ, NARRATE, & DISCUSS**
Part I: Introduction & The Sun

**→ PREP**
● Review where the four cardinal directions point in your place.
● Practice measuring objects at a distance using your hands.
● You will view during the day this week - check the weather and decide when you will do some sky gazing. Will you view it at sunrise? Daytime? Sunset? Since most of the work in this course will be at night, make the most of this week!
● Ensure that you understand how to set up your binoculars securely, so that you can project the sun for safe viewing:
∞ Website Link: Track Activity on the Surface of the Sun
∞ Video Link: How to Project the Sun with Binoculars
● If you would like to consider an equinox special study, check out the idea below:
∞ Website Link: Sun Shadows

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 3 ⬚ **45m Astronomy Lessons - Lesson 12**
*How Astronomy Works, Spatial Relations*

> ▯ Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit I Lsn 4. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 3 ⬚ **45m Astronomy Lessons - Lesson 13**
*Astronomy Literature*

> ▯ Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 3 ⬚ **45m Astronomy Lessons - Lesson 14**
*Astronomy Literature*

> ▯ Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

# Science: Astronomy

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## Term 1

WEEK 3 ⬚ **45m Astronomy Lessons - Lesson 15**
*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 3 ⬚ **60m Astronomy Labs - Lesson 3**

Materials: journal, binoculars, paper, tripod/stand for binoculars

**→ OBSERVE**
Use your hands to measure locations and your binocular set-up to observe the sun's projection. DO NOT LOOK DIRECTLY AT THE SUN! Keep a record of what you observe.

WEEK 4 ⬚ **45m Astronomy Lessons - Lesson 16**
*Identification*

Materials: 100 Things to See in the Night Sky

**→ READ, NARRATE, & DISCUSS**
The Moon

**→ PREP**
- Practice measuring objects at a distance using your hands.
- Review your previous nighttime viewing experience. Can you do anything to improve your experience?
- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Check the weather and note which nights will have cloud cover so that you can plan a night week to do some sky gazing.
- Note when the moon will rise and set this week to plan for viewing.
- If you would like to consider a lunar special study, you might journal the phases over the next month. For a more challenging study, check out the idea below:

∞ Link: Isabel Williamson Lunar Observing Program

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 4 ⬚ **45m Astronomy Lessons - Lesson 17**
*How Astronomy Works, Modern Astronomy*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit I Lsn 5. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

# Science: Astronomy

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## Term 1

**WEEK 4** ⬚ **45m Astronomy Lessons - Lesson 18**
*Astronomy Literature*

> Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

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**WEEK 4** ⬚ **45m Astronomy Lessons - Lesson 19**
*Astronomy Literature*

> Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

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**WEEK 4** ⬚ **45m Astronomy Lessons - Lesson 20**
*Astronomy Current Events and Research*

> Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

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**WEEK 4** ⬚ **60m Astronomy Labs - Lesson 4**

> Materials: journal, star finder, binoculars/telescope

**→ OBSERVE**
Use your hands to measure locations of the moon and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope to view the moon up close. Keep a record of what you observe.

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**WEEK 5** ⬚ **45m Astronomy Lessons - Lesson 21**
⬚ *45m Astronomy - Lesson 21*

> Materials: 100 Things to See in the Night Sky

Identification

**→ READ, NARRATE, & DISCUSS**
The Naked-Eye Planets & Mercury

**→ PREP**
- Practice measuring objects at a distance using your hands.
- Use the Stellarium and Night Sky Quick Links that tell you what is visible

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

# Science: Astronomy

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## Term 1

to aid your viewing. Depending on which planets will be visible this month, you may want to reorder your readings. For example, if Jupiter is visible next week, then consider reading about Jupiter next week and come back to the others later.
• Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 22**
*How Astronomy Works, Electromagnetic Waves*

Materials: Astronomy Text

**➜ READ, NARRATE, & DISCUSS**
Unit I Lsn 6. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 23**
*Astronomy Literature*

Materials: The Last Stargazers

**➜ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**➜ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 24**
*Astronomy Literature*

Materials: The Last Stargazers

**➜ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**➜ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 25**
*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**➜ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**➜ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

# Science: Astronomy

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## Term 1

WEEK 5

⬚ **60m Astronomy Labs - Lesson 5**

▯ Materials: journal, star finder, binoculars/telescope

**→ OBSERVE**
Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.

WEEK 6

⬚ **45m Astronomy Lessons - Lesson 26**
*Identification*

▯ Materials: 100 Things to See in the Night Sky

**→ READ, NARRATE, & DISCUSS**
Venus

**→ PREP**
• Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. Depending on which planets will be visible this month, you may want to reorder your readings.
• Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 6

⬚ **45m Astronomy Lessons - Lesson 27**
*How Astronomy Works, Frequency and Wavelength*

▯ Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit I Lsn 7. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 6

⬚ **45m Astronomy Lessons - Lesson 28**
*Astronomy Literature*

▯ Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 6

⬚ **45m Astronomy Lessons - Lesson 29**
*Astronomy Literature*

▯ Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

# Science: Astronomy

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## Term 1

WEEK 6 ⬚ **45m Astronomy Lessons - Lesson 30**
*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 6 ⬚ **60m Astronomy Labs - Lesson 6**

Materials: journal, star finder, binoculars/telescope

**→ OBSERVE**
Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.

WEEK 7 ⬚ **45m Astronomy Lessons - Lesson 31**
*Identification*

Materials: 100 Things to See in the Night Sky

**→ READ, NARRATE, & DISCUSS**
Mars

**→ PREP**
- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. Depending on which planets will be visible this month, you may want to reorder your readings.
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

- **PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 7 ⬚ **45m Astronomy Lessons - Lesson 32**
*How Astronomy Works, Spectral Analysis*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit I Lsn 8. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 7 ⬚ **45m Astronomy Lessons - Lesson 33**
*Astronomy Literature*

Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

# Science: Astronomy

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## Term 1

**→ WRITE**
Write a narration in your Astronomy Notebook.

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WEEK 7

⬚ **45m Astronomy Lessons - Lesson 34**
*Astronomy Literature*

> Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

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WEEK 7

⬚ **45m Astronomy Lessons - Lesson 35**
*Astronomy Current Events and Research*

> Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

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WEEK 7

⬚ **60m Astronomy Labs - Lesson 7**

> Materials: journal, star finder, binoculars/telescope

**→ OBSERVE**
Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.

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WEEK 8

⬚ **45m Astronomy Lessons - Lesson 36**
*Identification*

> Materials: 100 Things to See in the Night Sky

**→ READ, NARRATE, & DISCUSS**
Jupiter

**→ PREP**
• Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. Depending on which planets will be visible this month, you may want to reorder your readings.
• Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

# Science: Astronomy

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## Term 1

WEEK 8 ⬚ **45m Astronomy Lessons - Lesson 37**
*How Astronomy Works, The Grand and Magnificent Universe*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit I Lsn 9. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 8 ⬚ **45m Astronomy Lessons - Lesson 38**
*Astronomy Literature*

Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 8 ⬚ **45m Astronomy Lessons - Lesson 39**
*Astronomy Literature*

Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 8 ⬚ **45m Astronomy Lessons - Lesson 40**
*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 8 ⬚ **60m Astronomy Labs - Lesson 8**

Materials: journal, star finder, binoculars/telescope

**→ OBSERVE**
Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.

# Science: Astronomy

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## Term 1

**WEEK 9** ⬚ **45m Astronomy Lessons - Lesson 41**
*Identification*

Materials: 100 Things to See in the Night Sky

**➔ READ, NARRATE, & DISCUSS**
Saturn

**➔ PREP**
● Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing. Depending on which planets will be visible this month, you may want to reorder your readings.
● Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

**WEEK 9** ⬚ **45m Astronomy Lessons - Lesson 42**
*How Astronomy Works, The Grand Design*

Materials: Astronomy Text

**➔ READ, NARRATE, & DISCUSS**
Unit I Lsn 10. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**WEEK 9** ⬚ **45m Astronomy Lessons - Lesson 43**
*Astronomy Literature*

Materials: The Last Stargazers

**➔ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**➔ WRITE**
Write a narration in your Astronomy Notebook.

**WEEK 9** ⬚ **45m Astronomy Lessons - Lesson 44**
*Astronomy Literature*

Materials: The Last Stargazers

**➔ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**➔ WRITE**
Write a narration in your Astronomy Notebook.

**WEEK 9** ⬚ **45m Astronomy Lessons - Lesson 45**
*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**➔ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**➔ PRACTICE CITIZEN SCIENCE**

# Science: Astronomy

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## Term 1

Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

---

WEEK 9

⬚ **60m Astronomy Labs - Lesson 9**

> Materials: journal, star finder, binoculars/telescope

**→ OBSERVE**
Use your hands to measure locations of the moon, planets, and other objects noted with your star finder and/or Quick Links. Use your binoculars/telescope, as desired, to magnify. Keep a record of what you observe.

---

WEEK 10

⬚ **45m Astronomy Lessons - Lesson 46**
*Identification*

> Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Part II: Introduction, The Northern Sky, & Ursa Major

**→ PREP**
● Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
● Use the star finder to note where Ursa Major will be.
● Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

---

WEEK 10

⬚ **45m Astronomy Lessons - Lesson 47**
*How Astronomy Works, The Telescope*

> Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit I Lsn 11. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

---

WEEK 10

⬚ **45m Astronomy Lessons - Lesson 48**
*Astronomy Literature*

> Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

---

WEEK 10

⬚ **45m Astronomy Lessons - Lesson 49**
*Astronomy Literature*

> Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

# Science: Astronomy

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## Term 1

WEEK 10 ⬚ **45m Astronomy Lessons - Lesson 50**
*Astronomy Current Events and Research*

> Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

---

WEEK 10 ⬚ **60m Astronomy Labs - Lesson 10**

> Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of Ursa Major, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

---

WEEK 11 ⬚ **45m Astronomy Lessons - Lesson 51**
*Identification*

> Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Merak and Dubhe & Mizar and Alcor

**→ PREP**
- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Use the star finder to note the position of Ursa Major and the associated stars.
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

---

WEEK 11 ⬚ **45m Astronomy Lessons - Lesson 52**
*How Astronomy Works, Space Exploration*

> Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit I Lsn 12. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

---

WEEK 11 ⬚ **45m Astronomy Lessons - Lesson 53**
*Astronomy Catch-up Day*

**→ CATCH-UP**
Use this time to complete any readings, note-booking, or projects for the course.

# Science: Astronomy

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## Term 1

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|---|---|---|
| WEEK 11 | ⬚ **45m Astronomy Lessons - Lesson 54**<br>*Astronomy Catch-up Day*<br><br>**→ CATCH-UP**<br>Use this time to complete any readings, note-booking, or projects for the course. | |
| WEEK 11 | ⬚ **45m Astronomy Lessons - Lesson 55**<br>*Astronomy Current Events and Research*<br><br>Materials: computer, optional: Crossroads of Science and Faith text<br><br>**→ READ CURRENT EVENTS**<br>EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.<br><br>OR<br><br>**→ PRACTICE CITIZEN SCIENCE**<br>Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do. | |
| WEEK 11 | ⬚ **60m Astronomy Labs - Lesson 11**<br><br>Materials: journal, star finder<br><br>**→ OBSERVE**<br>Use your hands to measure locations of Ursa Major and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe. | |
| WEEK 12 | ⬚ **45m Astronomy Lessons - Lesson 56**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to the course. | |
| WEEK 12 | ⬚ **45m Astronomy Lessons - Lesson 57**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to the course. | |
| WEEK 12 | ⬚ **45m Astronomy Lessons - Lesson 58**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to the course. | |
| WEEK 12 | ⬚ **45m Astronomy Lessons - Lesson 59**<br>*Exam Week*<br><br>**→ EXAMS**<br>Answer question(s) related to the course. | |

# Science: Astronomy

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## Term 1

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|---|---|---|
| WEEK 12 | ☐ **45m Astronomy Lessons - Lesson 60**<br>*Exam Week*<br><br>**➔ EXAMS**<br>Answer question(s) related to the course. | |
| WEEK 12 | ☐ **60m Astronomy Labs - Lesson 12**<br>*Exam Week*<br><br>**➔ EXAMS**<br>Answer question(s) related to the course. | |

# Science: Astronomy

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## Term 2

**• PLAN WEEKLY**
- [ ] nature walk - record observations
- [ ] science free read

WEEK 1 [ ] **45m Astronomy Lessons - Lesson 61**
*Identification*

Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Ursa Minor, Polaris, & Kochab and Pherkad

**→ PREP**
• Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
• Use the star finder to note the position of Ursa Minor and the associated stars.
• Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

WEEK 1 [ ] **45m Astronomy Lessons - Lesson 62**
*How Astronomy Works, The Solar System*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit II Lsn 1. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 1 [ ] **45m Astronomy Lessons - Lesson 63**
*Astronomy Literature*

Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 1 [ ] **45m Astronomy Lessons - Lesson 64**
*Astronomy Literature*

Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 1 [ ] **45m Astronomy Lessons - Lesson 65**
*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**→ INTRO**
Use this time to explore current events in astronomy, keeping a weekly log or journal entry of your findings and thoughts in your Astronomy Notebook.

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of

# Science: Astronomy

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## Term 2

Science and Faith text could be substituted here.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links).

---

WEEK 1

☐ **60m Astronomy Labs - Lesson 13**

Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of Ursa Major, Ursa Minor, and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Has the location of Ursa Major changed from your last viewing? Do the locations change during your session? Keep a record of what you observe.

---

WEEK 2

☐ **45m Astronomy Lessons - Lesson 66**
*Identification*

Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Cassiopeia & Caph

**→ PREP**
- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Use the star finder to note the position of Cassiopeia and its associated star(s).
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

- **PLAN WEEKLY**
☐ nature walk - record observations
☐ science free read

---

WEEK 2

☐ **45m Astronomy Lessons - Lesson 67**
*How Astronomy Works, Formation of Our Solar System*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit II Lsn 2. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

---

WEEK 2

☐ **45m Astronomy Lessons - Lesson 68**
*Astronomy Literature*

Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

---

WEEK 2

☐ **45m Astronomy Lessons - Lesson 69**
*Astronomy Literature*

Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

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## Term 2

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 2 ⬚ **45m Astronomy Lessons - Lesson 70**
*Astronomy Current Events and Research*

> Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 2 ⬚ **60m Astronomy Labs - Lesson 14**

> Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 3 ⬚ **45m Astronomy Lessons - Lesson 71**
*Identification*

> Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Cepheus & Delta Cephei

**→ PREP**
- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Use the star finder to note the position of Cepheus and its associated star(s).
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 3 ⬚ **45m Astronomy Lessons - Lesson 72**
*How Astronomy Works, The Sun*

> Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit II Lsn 3. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
Explore the Sun at NASA:
∞ Link: NASA - Sun

# Science: Astronomy

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## Term 2

WEEK 3 ⬚ **45m Astronomy Lessons - Lesson 73**
*Astronomy Literature*

▯ Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 3 ⬚ **45m Astronomy Lessons - Lesson 74**
*Astronomy Literature*

▯ Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 3 ⬚ **45m Astronomy Lessons - Lesson 75**
*Astronomy Current Events and Research*

▯ Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 3 ⬚ **60m Astronomy Labs - Lesson 15**

▯ Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 4 ⬚ **45m Astronomy Lessons - Lesson 76**
*Identification*

▯ Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Draco & Thuban

**→ PREP**
• Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
• Use the star finder to note the position of Draco and its associated

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

# Science: Astronomy

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## Term 2

star(s).
• Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

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WEEK 4 ⬚ **45m Astronomy Lessons - Lesson 77**
*How Astronomy Works, Earth & Moon*

> Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit II Lsn 4-5. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
Explore Earth and the Moon at NASA:
∞ Link: NASA - Solar System

---

WEEK 4 ⬚ **45m Astronomy Lessons - Lesson 78**
*Astronomy Literature*

> Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

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WEEK 4 ⬚ **45m Astronomy Lessons - Lesson 79**
*Astronomy Literature*

> Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

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WEEK 4 ⬚ **45m Astronomy Lessons - Lesson 80**
*Astronomy Current Events and Research*

> Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

# Science: Astronomy

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## Term 2

WEEK 4 ⬚ **60m Astronomy Labs - Lesson 16**

▯ Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 81**
*Identification*

▯ Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Now move to whatever season you are in and continue reading.

**→ PREP**
● Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
● Use the star finder to note the position of the stars you are interested in.
● Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 82**
*How Astronomy Works, Comets*

▯ Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit II Lsn 6. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
Explore comets at NASA:
∞ Link: NASA - Solar System

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 83**
*Astronomy Literature*

▯ Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 84**
*Astronomy Literature*

▯ Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

# Science: Astronomy

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## Term 2

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 85**
*Astronomy Current Events and Research*

> Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

---

WEEK 5 ⬚ **60m Astronomy Labs - Lesson 17**

> Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

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WEEK 6 ⬚ **45m Astronomy Lessons - Lesson 86**
*Identification*

> Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Use the star finder to note the position of the stars you are interested in.
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

---

WEEK 6 ⬚ **45m Astronomy Lessons - Lesson 87**
*How Astronomy Works, Asteroids and Meteors*

> Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit II Lsn 7 - half of Lsn 8. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
Explore asteroids at NASA:
∞ Link: NASA - Solar System

# Science: Astronomy

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## Term 2

**WEEK 6** ⬚ **45m Astronomy Lessons - Lesson 88**
*Astronomy Literature*

> Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

---

**WEEK 6** ⬚ **45m Astronomy Lessons - Lesson 89**
*Astronomy Literature*

> Materials: The Last Stargazers

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

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**WEEK 6** ⬚ **45m Astronomy Lessons - Lesson 90**
*Astronomy Current Events and Research*

> Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

---

**WEEK 6** ⬚ **60m Astronomy Labs - Lesson 18**

> Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

---

**WEEK 7** ⬚ **45m Astronomy Lessons - Lesson 91**
*Identification*

> Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
• Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.

• **PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

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## Term 2

● Use the star finder to note the position of the stars you are interested in.
● Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

WEEK 7 ⬚ **45m Astronomy Lessons - Lesson 92**
*How Astronomy Works, Meteors and Impactors*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit II, the remaining half of Lsn 8 - Lsn 9. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
Explore meteors at NASA:
∞ Link: NASA - Solar System

WEEK 7 ⬚ **45m Astronomy Lessons - Lesson 93**
*Astronomy Literature*

Materials: The Last Stargazers, optional: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days). When finished with The Last Stargazers, begin The Day Without Yesterday.

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 7 ⬚ **45m Astronomy Lessons - Lesson 94**
*Astronomy Literature*

Materials: The Last Stargazers, optional: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days). When finished with The Last Stargazers, begin The Day Without Yesterday.

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 7 ⬚ **45m Astronomy Lessons - Lesson 95**
*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

  

# Science: Astronomy

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## Term 2

WEEK 7

⬚ **60m Astronomy Labs - Lesson 19**

Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 8

⬚ **45m Astronomy Lessons - Lesson 96**
*Identification*

Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
● Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
● Use the star finder to note the position of the stars you are interested in.
● Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 8

⬚ **45m Astronomy Lessons - Lesson 97**
*How Astronomy Works, Highlights of the Planets*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit II Lsn 10. If you want to break this chapter up, feel free to stop when full; you may pick up next week. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
You will explore the most up-to-date knowledge about the planets for the rest of the term at NASA:
∞ Link: NASA - Solar System

WEEK 8

⬚ **45m Astronomy Lessons - Lesson 98**
*Astronomy Literature*

Materials: The Last Stargazers, optional: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days). When finished with The Last Stargazers, begin The Day Without Yesterday.

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 8

⬚ **45m Astronomy Lessons - Lesson 99**
*Astronomy Literature*

Materials: The Last Stargazers, optional: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 19 pages per week (over 2 lesson days). When finished with The Last Stargazers, begin The Day Without Yesterday.

# Science: Astronomy

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## Term 2

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 8

⬚ **45m Astronomy Lessons - Lesson 100**
*Astronomy Current Events and Research*

▯ Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 8

⬚ **60m Astronomy Labs - Lesson 20**

▯ Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 9

⬚ **45m Astronomy Lessons - Lesson 101**
*Identification*

▯ Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
● Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
● Use the star finder to note the position of the stars you are interested in.
● Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 9

⬚ **45m Astronomy Lessons - Lesson 102**
*How Astronomy Works, The Planets Continued*

▯ Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Continue Unit II Lsn 10. as needed. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
You will explore the most up-to-date knowledge about the planets for the rest of the term at NASA:
∞ Link: NASA - Solar System

  

# Science: Astronomy

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## Term 2

**WEEK 9** ⬚ **45m Astronomy Lessons - Lesson 103**
*Astronomy Literature*

▯ Materials: The Day Without Yesterday

**➔ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days). There will be some math and science in the story that you have not yet learned, but that is okay. The point is really about the story.

**➔ WRITE**
Write a narration in your Astronomy Notebook.

**WEEK 9** ⬚ **45m Astronomy Lessons - Lesson 104**
*Astronomy Literature*

▯ Materials: The Day Without Yesterday

**➔ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**➔ WRITE**
Write a narration in your Astronomy Notebook.

**WEEK 9** ⬚ **45m Astronomy Lessons - Lesson 105**
*Astronomy Current Events and Research*

▯ Materials: computer, optional: Crossroads of Science and Faith text

**➔ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**➔ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

**WEEK 9** ⬚ **60m Astronomy Labs - Lesson 21**

▯ Materials: journal, star finder

**➔ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

**WEEK 10** ⬚ **45m Astronomy Lessons - Lesson 106**
*Identification*

▯ Materials: 100 Things to See in the Night Sky, star finder

**➔ READ, NARRATE, & DISCUSS**
Continue reading.

**➔ PREP**
● Use the Stellarium and Night Sky Quick Links that tell you what is visible

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

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to aid your viewing.
• Use the star finder to note the position of the stars you are interested in.
• Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

WEEK 10 **45m Astronomy Lessons - Lesson 107**
*How Astronomy Works, The Planets Continued*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Continue Unit II Lsn 10. as needed. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
You will explore the most up-to-date knowledge about the planets for the rest of the term at NASA:
∞ Link: NASA - Solar System

WEEK 10 **45m Astronomy Lessons - Lesson 108**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 10 **45m Astronomy Lessons - Lesson 109**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 10 **45m Astronomy Lessons - Lesson 110**
*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

  

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WEEK 10 ⬚ **60m Astronomy Labs - Lesson 22**

▯ Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 11 ⬚ **45m Astronomy Lessons - Lesson 111**
*Identification*

▯ Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
● Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
● Use the star finder to note the position of the stars you are interested in.
● Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

**● PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 11 ⬚ **45m Astronomy Lessons - Lesson 112**
*How Astronomy Works, The Planets Continued*

▯ Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Continue Unit II Lsn 10. as needed. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
You will explore the most up-to-date knowledge about the planets for the rest of the term at NASA:
∞ Link: NASA - Solar System

WEEK 11 ⬚ **45m Astronomy Lessons - Lesson 113**
*Astronomy Catch-up Day*

**→ CATCH-UP**
Use this time to complete any readings, note-booking, or projects for the course.

WEEK 11 ⬚ **45m Astronomy Lessons - Lesson 114**
*Astronomy Catch-up Day*

**→ CATCH-UP**
Use this time to complete any readings, note-booking, or projects for the course.

WEEK 11 ⬚ **45m Astronomy Lessons - Lesson 115**
*Astronomy Current Events and Research*

▯ Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of

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Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 11 ⬚ **60m Astronomy Labs - Lesson 23**

Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 12 ⬚ **45m Astronomy Lessons - Lesson 116**
*Exam Week*

**→ EXAMS**
Answer question(s) related to the course.

WEEK 12 ⬚ **45m Astronomy Lessons - Lesson 117**
*Exam Week*

**→ EXAMS**
Answer question(s) related to the course.

WEEK 12 ⬚ **45m Astronomy Lessons - Lesson 118**
*Exam Week*

**→ EXAMS**
Answer question(s) related to the course.

WEEK 12 ⬚ **45m Astronomy Lessons - Lesson 119**
*Exam Week*

**→ EXAMS**
Answer question(s) related to the course.

WEEK 12 ⬚ **45m Astronomy Lessons - Lesson 120**
*Exam Week*

**→ EXAMS**
Answer question(s) related to the course.

WEEK 12 ⬚ **60m Astronomy Labs - Lesson 24**
*Exam Week*

**→ EXAMS**
Answer question(s) related to the course.

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## Term 3

**WEEK 1** ⬚ **45m Astronomy Lessons - Lesson 121**
*Identification*

Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
• Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
• Use the star finder to note the position of the stars you are interested in.
• Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

**WEEK 1** ⬚ **45m Astronomy Lessons - Lesson 122**
*How Astronomy Works, Cosmology*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit III Lsn 1. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**WEEK 1** ⬚ **45m Astronomy Lessons - Lesson 123**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

**WEEK 1** ⬚ **45m Astronomy Lessons - Lesson 124**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

**WEEK 1** ⬚ **45m Astronomy Lessons - Lesson 125**
*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**→ READ**
Use this time to explore current events in astronomy, keeping a weekly log or journal entry of your findings and thoughts in your Astronomy Notebook.

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here.

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## Term 3

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links).

WEEK 1 ⬚ **60m Astronomy Labs - Lesson 25**

Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 2 ⬚ **45m Astronomy Lessons - Lesson 126**
*Identification*

Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
• Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
• Use the star finder to note the position of the stars you are interested in.
• Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 2 ⬚ **45m Astronomy Lessons - Lesson 127**
*How Astronomy Works, Cosmology, and Theology*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit III Lsn 2. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

WEEK 2 ⬚ **45m Astronomy Lessons - Lesson 128**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 2 ⬚ **45m Astronomy Lessons - Lesson 129**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

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**WEEK 2**

☐ **45m Astronomy Lessons - Lesson 130**

*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

**WEEK 2**

☐ **60m Astronomy Labs - Lesson 26**

Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

**WEEK 3**

☐ **45m Astronomy Lessons - Lesson 131**

*Identification*

Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
- Use the Stellarium and Night Sky Quick Links that tell you what is visible to aid your viewing.
- Use the star finder to note the position of the stars you are interested in.
- Check the weather and note which nights will have cloud cover so that you can plan a night to do some sky gazing.

- **PLAN WEEKLY**

☐ nature walk - record observations
☐ science free read

**WEEK 3**

☐ **45m Astronomy Lessons - Lesson 132**

*How Astronomy Works, Beyond the Solar System*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit III half of Lsn 3. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
Explore images from the Hubble and Webb telescopes in your Quick Links, as desired.

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## Term 3

WEEK 3 ⬚ **45m Astronomy Lessons - Lesson 133**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 3 ⬚ **45m Astronomy Lessons - Lesson 134**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 3 ⬚ **45m Astronomy Lessons - Lesson 135**
*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 3 ⬚ **60m Astronomy Labs - Lesson 27**

Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 4 ⬚ **45m Astronomy Lessons - Lesson 136**
*Identification*

Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
Use your Quick Links, star finder, and weather forecast to plan for sky gazing.

• **PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

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WEEK 4 ⬚ **45m Astronomy Lessons - Lesson 137**
*How Astronomy Works, Beyond the Solar System*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit III, the remaining half of Lsn 3. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
Explore images from the Hubble and Webb telescopes in your Quick Links, as desired.

WEEK 4 ⬚ **45m Astronomy Lessons - Lesson 138**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 4 ⬚ **45m Astronomy Lessons - Lesson 139**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 4 ⬚ **45m Astronomy Lessons - Lesson 140**
*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 4 ⬚ **60m Astronomy Labs - Lesson 28**

Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated

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stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 141**
*Identification*

▯ Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
Use your Quick Links, star finder, and weather forecast to plan for sky gazing.

• **PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 142**
*How Astronomy Works, The Life Cycle of a Star*

▯ Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit III, half of Lsn 4. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
Explore images from the Hubble and Webb telescopes in your Quick Links, as desired.

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 143**
*Astronomy Literature*

▯ Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 144**
*Astronomy Literature*

▯ Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 5 ⬚ **45m Astronomy Lessons - Lesson 145**
*Astronomy Current Events and Research*

▯ Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

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OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

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**WEEK 5** ⬚ **60m Astronomy Labs - Lesson 29**

Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

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**WEEK 6** ⬚ **45m Astronomy Lessons - Lesson 146**
*Identification*

Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
Use your Quick Links, star finder, and weather forecast to plan for sky gazing.

**• PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

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**WEEK 6** ⬚ **45m Astronomy Lessons - Lesson 147**
*How Astronomy Works, The Life Cycle of a Star*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit III, the remaining half of Lsn 4. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
Explore images from the Hubble and Webb telescopes in your Quick Links, as desired.

---

**WEEK 6** ⬚ **45m Astronomy Lessons - Lesson 148**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

---

**WEEK 6** ⬚ **45m Astronomy Lessons - Lesson 149**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

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**→ WRITE**
Write a narration in your Astronomy Notebook.

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WEEK 6

⬚ **45m Astronomy Lessons - Lesson 150**
*Astronomy Current Events and Research*

> ▯ Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

---

WEEK 6

⬚ **60m Astronomy Labs - Lesson 30**

> ▯ Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

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WEEK 7

⬚ **45m Astronomy Lessons - Lesson 151**
*Identification*

> ▯ Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
Use your Quick Links, star finder, and weather forecast to plan for sky gazing.

• **PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

---

WEEK 7

⬚ **45m Astronomy Lessons - Lesson 152**
*How Astronomy Works, Intrinsic Variables, and Determining Distance*

> ▯ Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit III Lsn 5. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
Explore images from the Hubble and Webb telescopes in your Quick Links, as desired.

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WEEK 7 ⬚ **45m Astronomy Lessons - Lesson 153**
*Astronomy Literature*

▯ Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

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WEEK 7 ⬚ **45m Astronomy Lessons - Lesson 154**
*Astronomy Literature*

▯ Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

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WEEK 7 ⬚ **45m Astronomy Lessons - Lesson 155**
*Astronomy Current Events and Research*

▯ Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

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WEEK 7 ⬚ **60m Astronomy Labs - Lesson 31**

▯ Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

---

WEEK 8 ⬚ **45m Astronomy Lessons - Lesson 156**
*Identification*

▯ Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
Use your Quick Links, star finder, and weather forecast to plan for sky gazing.

• **PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

# Science: Astronomy

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## Term 3

WEEK 8 ⬚ **45m Astronomy Lessons - Lesson 157**
*How Astronomy Works, Ways to Determine Distances*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit III Lsn 6. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ VIEW**
Explore images from the Hubble and Webb telescopes in your Quick Links, as desired.

WEEK 8 ⬚ **45m Astronomy Lessons - Lesson 158**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 8 ⬚ **45m Astronomy Lessons - Lesson 159**
*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

WEEK 8 ⬚ **45m Astronomy Lessons - Lesson 160**
*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

WEEK 8 ⬚ **60m Astronomy Labs - Lesson 32**

Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated

Click THIS text or scan the QR code for links.

## Term 3

stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

**• PLAN WEEKLY**
- [ ] nature walk - record observations
- [ ] science free read

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WEEK 9

- [ ] **45m Astronomy Lessons - Lesson 161**

*Identification*

Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
Use your Quick Links, star finder, and weather forecast to plan for sky gazing.

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WEEK 9

- [ ] **45m Astronomy Lessons - Lesson 162**

*How Astronomy Works, The H-R Diagram*

Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit III Lsn 7. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ PROJECT**
Use any additional time to work on any projects (e.g., special studies, citizen science, etc.).

---

WEEK 9

- [ ] **45m Astronomy Lessons - Lesson 163**

*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

---

WEEK 9

- [ ] **45m Astronomy Lessons - Lesson 164**

*Astronomy Literature*

Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

---

WEEK 9

- [ ] **45m Astronomy Lessons - Lesson 165**

*Astronomy Current Events and Research*

Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

  

## Term 3

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

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WEEK 9 ⬚ **60m Astronomy Labs - Lesson 33**

▯ Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

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WEEK 10 ⬚ **45m Astronomy Lessons - Lesson 166**
*Identification*

▯ Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
Use your Quick Links, star finder, and weather forecast to plan for sky gazing.

• **PLAN WEEKLY**
⬚ nature walk - record observations
⬚ science free read

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WEEK 10 ⬚ **45m Astronomy Lessons - Lesson 167**
*How Astronomy Works, Other Planetary Systems*

▯ Materials: Astronomy Text

**→ READ, NARRATE, & DISCUSS**
Unit III Lsn 8. Narrate silently to yourself or write. Copy any diagrams and make sketches if helpful to understand the text. Consider any questions from the text as time permits.

**→ PROJECT**
Use any additional time to work on any projects (e.g., special studies, citizen science, etc.).

---

WEEK 10 ⬚ **45m Astronomy Lessons - Lesson 168**
*Astronomy Literature*

▯ Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

**→ WRITE**
Write a narration in your Astronomy Notebook.

---

WEEK 10 ⬚ **45m Astronomy Lessons - Lesson 169**
*Astronomy Literature*

▯ Materials: The Day Without Yesterday

**→ READ, NARRATE, & DISCUSS**
Read about 18 pages per week (over 2 lesson days).

# Science: Astronomy

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## Term 3

**→ WRITE**
Write a narration in your Astronomy Notebook.

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WEEK 10 ⬚ **45m Astronomy Lessons - Lesson 170**
*Astronomy Current Events and Research*

> Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

---

WEEK 10 ⬚ **60m Astronomy Labs - Lesson 34**

> Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

---

WEEK 11 ⬚ **45m Astronomy Lessons - Lesson 171**
*Identification*

> Materials: 100 Things to See in the Night Sky, star finder

**→ READ, NARRATE, & DISCUSS**
Continue reading.

**→ PREP**
Use your Quick Links, star finder, and weather forecast to plan for sky gazing.

- **PLAN WEEKLY**
  ⬚ nature walk - record observations
  ⬚ science free read

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WEEK 11 ⬚ **45m Astronomy Lessons - Lesson 172**
*Astronomy Catch-up Day*

**→ CATCH-UP**
Use this time to complete any readings, note-booking, or projects for the course.

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WEEK 11 ⬚ **45m Astronomy Lessons - Lesson 173**
*Astronomy Catch-up Day*

**→ CATCH-UP**
Use this time to complete any readings, note-booking, or projects for the course.

# Science: Astronomy

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## Term 3

**WEEK 11** ⬚ **45m Astronomy Lessons - Lesson 174**
*Astronomy Catch-up Day*

**→ CATCH-UP**
Use this time to complete any readings, note-booking, or projects for the course.

**WEEK 11** ⬚ **45m Astronomy Lessons - Lesson 175**
*Astronomy Current Events and Research*

> Materials: computer, optional: Crossroads of Science and Faith text

**→ READ CURRENT EVENTS**
EarthSky, Picture of Day, and Veritas Forum (in your Quick Links) are good options to explore. If you and your teacher prefer, the Crossroads of Science and Faith text could be substituted here. Make an entry in your notebook about what you discover.

OR

**→ PRACTICE CITIZEN SCIENCE**
Choose from Galaxy Zoo, Globe at Night, Exoplanet Watch, or Space Math (in your Quick Links). Log an entry in your notebook about what you do.

**WEEK 11** ⬚ **60m Astronomy Labs - Lesson 35**

> Materials: journal, star finder

**→ OBSERVE**
Use your hands to measure locations of constellations and associated stars, as well as the moon, planets, and other objects noted with your star finder and/or Quick Links. Keep a record of what you observe.

**WEEK 12** ⬚ **45m Astronomy Lessons - Lesson 176**
*Exam Week*

**→ EXAMS**
Answer question(s) related to the course.

**WEEK 12** ⬚ **45m Astronomy Lessons - Lesson 177**
*Exam Week*

**→ EXAMS**
Answer question(s) related to the course.

**WEEK 12** ⬚ **45m Astronomy Lessons - Lesson 178**
*Exam Week*

**→ EXAMS**
Answer question(s) related to the course.

**WEEK 12** ⬚ **45m Astronomy Lessons - Lesson 179**
*Exam Week*

**→ EXAMS**
Answer question(s) related to the course.

# Science: Astronomy

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## Term 3

| | | |
|---|---|---|
| WEEK 12 | ⬚ **45m Astronomy Lessons - Lesson 180**<br>*Exam Week*<br><br>**➔ EXAMS**<br>Answer question(s) related to the course. | |
| WEEK 12 | ⬚ **60m Astronomy Labs - Lesson 36**<br>*Exam Week*<br><br>**➔ EXAMS**<br>Answer question(s) related to the course. | |

# Science: Astronomy

## Examination

### Term 1

**Astronomy Lessons**

· 100 Things to See: Tell something you learned about observing the sun, moon, or planets. OR Tell about stargazing this term. For either question, share your notebook. What did you observe? What questions did you think about?

· Astronomy: Explain how we can know that something exists if we cannot see it (such as wind and black holes)?

· Astronomy: Why is light important to astronomy? What is light? What is the Doppler effect?

· The Last Stargazers: Tell about something that surprised you about the life and work of an astronomer from what you have read so far. OR, Many people see science and art as very different pursuits, but the author tells about scientists who love music, poetry, drawing, etc. Explain how you think science and art work together as a person explores the Things around them.

· Tell about a current event or citizen science project from this term.

### Term 2

**Astronomy Lessons**

· 100 Things to See: Explain how to find some of the major constellations, such as Ursa Major/minor, Cassiopeia, Draco, etc. OR Tell about stargazing this term. For either question, share your notebook. What did you observe? What questions did you think about?

· Astronomy: Tell about 3 unique or interesting features found among objects in the solar system.

· The Last Stargazers: Tell about a scientific aspect of astronomy (the discoveries, the mysteries, the science of telescopes, etc) that you found interesting.

· The Day Without Yesterday: The scientific community persistently resisted the idea of an expanding universe. Give reasons for this from what you read.

· Tell about a current event or citizen science project from this term.

### Term 3

**Astronomy Lessons**

· 100 Things to See: Tell about some of the seasonal stars you learned to find this term. OR Tell about stargazing this term. For either question, share your notebook. What did you observe? What questions did you think about?

· Astronomy: Explain what a galaxy, black hole, nebula, supernova, and binary star system is? How are they connected? OR Explain 4 ideas as to why the universe seems designed for life.

· Astronomy: Describe the life cycle of a star. OR What are some questions that theology and cosmology share?

· The Day Without Yesterday: Lemaître's work has been described as 'apolegetics in action.' Explain.

· Tell about a current event or citizen science project from this term.